

HARMONIC OSCILLATOR MODELS OF DARK MATTER

Unifying Ultralight Dark Matter Models with the Harmonic Framework's T_3 Branch

Author: Peter James Thompson

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ABSTRACT

Ultralight dark matter models treat dark matter as a coherently oscillating classical field — a harmonic oscillator. The Harmonic Framework of Reality posits that dark matter is the T_3 time dimension, with a mass ladder derived from the 27 Hz anchor. This paper demonstrates that the harmonic oscillator models of dark matter are exactly the T_3 branch of the Tau lattice.

Keywords: Dark matter, harmonic oscillator, T_3 branch, ultralight dark matter, 27 Hz anchor

1. INTRODUCTION

1.1 Ultralight Dark Matter

Ultralight dark matter (ULDM) models treat dark matter as a coherently oscillating scalar field:

$$\phi(t) = \phi_0 \times \cos(\omega t)$$

The mass is:

$$m = \hbar\omega/c^2$$

1.2 The Harmonic Framework's T_3 Branch

The Harmonic Framework posits that dark matter is the T_3 time dimension:

$$f_{c3} = 27 \times (13/19) / 230 = 0.08032 \text{ Hz}$$

The mass ladder is:

$$m_{dm} = (h \times f_{c3} / c^2) \times k$$

1.3 The Thesis

ULDM models are the T_3 branch. The harmonic oscillator frequency is:

$$\omega = 2\pi \times 0.08032 \text{ Hz} \times k$$

2. THE HARMONIC OSCILLATOR MAPPING

2.1 The ULDM Field

The ULDM scalar field is:

$$\varphi(t) = \sqrt{(2\rho) / m} \times \cos(m t)$$

2.2 The T₃ Branch

The T₃ branch is:

$$\Psi(t) = \sum_{k=1}^{32} A_k \times \cos(2\pi \times f_{c3} \times k \times t + \varphi_k)$$

The mass ladder is:

$$m_k = (h \times f_{c3} / c^2) \times k$$

2.3 The Mapping

The mapping is exact:

ULDM harmonic oscillator = T₃ branch harmonic mode

3. COMPARISON WITH OBSERVATIONS

3.1 Ultralight Dark Matter

For $k = 1$:

$$m = 3.31 \times 10^{-14} \text{ eV}$$

This is the ultralight dark matter range.

3.2 Sterile Neutrino Range

For $k = 10^{12}$:

$$m = 0.033 \text{ eV}$$

This is the sterile neutrino range.

3.3 WIMP Range

For $k = 10^{22}$:

$$m = 3.3 \times 10^8 \text{ eV} = 330 \text{ GeV}$$

This is the WIMP range.

4. CONCLUSION

4.1 Summary

Ultralight dark matter models are the T_3 branch. The harmonic oscillator frequency is derived from the 27 Hz anchor.

4.2 Testable Predictions

1. Dark matter should show the 32-harmonic comb
 2. The mass ladder follows $k \times f_c^3$
 3. The amplitude follows the 19:13:4 envelope
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Correspondence: peterjamesthompson101@gmail.com

Repository: <https://github.com/peterjamesthompson101-svg/Physics-Papers>

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